



A Level • Cambridge (CIE) • Chemistry

24 mins

13 questions

Multiple Choice Practice

Topical Practice MCQ

8. Reaction kinetics

Carefully selected past paper questions by topic.

15 The equations for two reactions are shown.



The two reactions have similar reaction mechanisms.

The initial rate of reaction X is greater than that of reaction Y when measured under identical conditions of temperature, pressure and reactant concentration.

Which statements explain this difference?

- 1 The activation energy for reaction X is less than that of reaction Y.
- 2 The Br–Br bond is weaker than the Cl–Cl bond.
- 3 A higher frequency of collisions between molecules of NOBr occur than between molecules of NOCl.

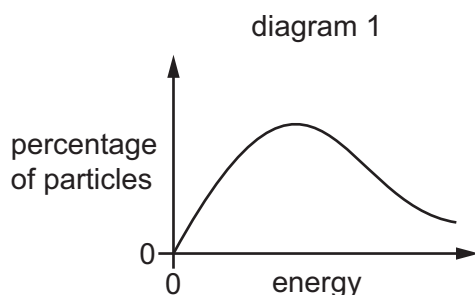
A 1 and 2

B 2 and 3

C 1 only

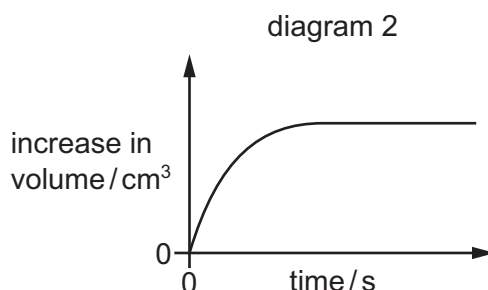
D 3 only

- 15 The Boltzmann distribution of the particles in a mixture of gas X and gas Y is shown in diagram 1.

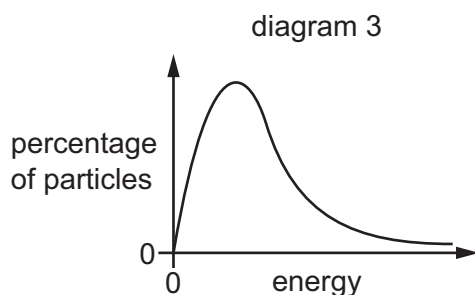


X and Y react and the reaction causes an increase in gas molecules present. The reaction goes to completion.

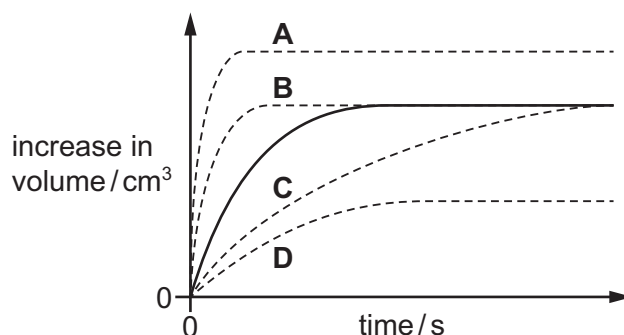
In experiment 1, the increase in volume is measured every 10 seconds. During the reaction, the temperature and pressure remain constant. The increase in volume is shown in the volume–time graph in diagram 2.



In experiment 2, the experiment is repeated using identical amounts of X and Y. A different temperature is used compared to experiment 1. The same pressure is used. The Boltzmann distribution of the second mixture of X and Y is shown in diagram 3. During the reaction the temperature and pressure remain constant.



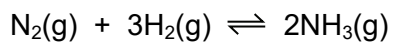
Which curve on the volume–time graph would show the increase in volume against time for experiment 2? (The original line for experiment 1 is redrawn as a solid line.)



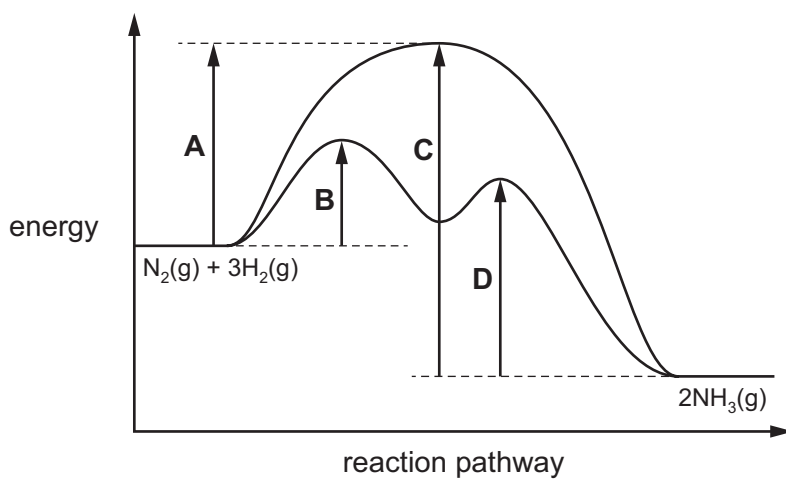
Question 3

9701_s21_qp_12 • Q7

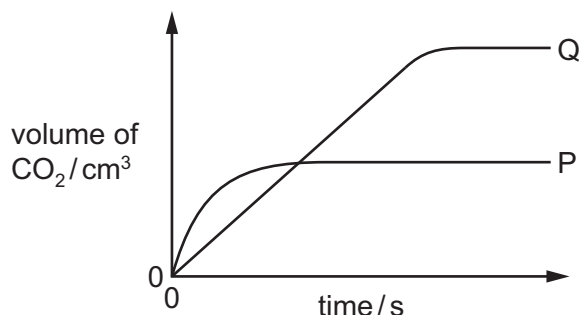
- 7 The reaction pathway diagram for the catalysed reaction and the uncatalysed reaction between N_2 and H_2 is shown.



Which letter represents the activation energy for the first step in the decomposition of NH_3 in the presence of a catalyst?



- 11 The volume of carbon dioxide collected by reacting 0.100 mol of magnesium carbonate with 50.0 cm³ of 1.00 mol dm⁻³ sulfuric acid is measured against time. The volume produced is plotted against time and the line labelled P on the graph is obtained.



The experiment is repeated using 0.100 mol of the same magnesium carbonate, and a different sample of acid. All other conditions remain the same. Plotting these results gives the line labelled Q.

Which sample of acid could give the line labelled Q?

- A 100 cm³ of 0.500 mol dm⁻³ sulfuric acid
- B 100 cm³ of 1.00 mol dm⁻³ sulfuric acid
- C 200 cm³ of 0.500 mol dm⁻³ hydrochloric acid
- D 200 cm³ of 1.00 mol dm⁻³ hydrochloric acid

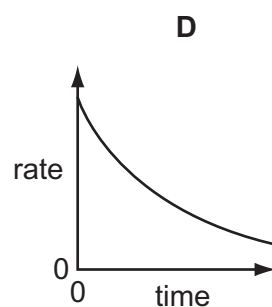
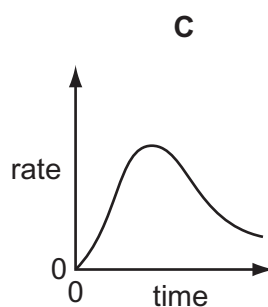
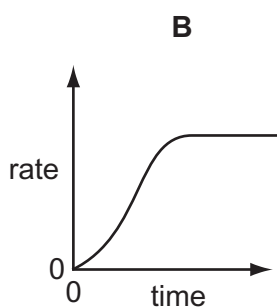
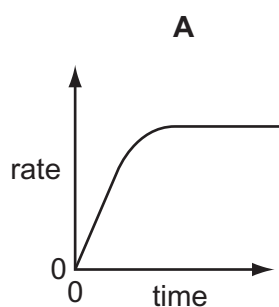
- 11** A chemical company used a catalyst in a chemical process. The company has now decided not to use the catalyst but to increase the temperature so that the rate of the reaction is the same as it was when the catalyst was used.

Which statement about the new conditions compared to the original conditions is correct?

- A** The activation energy has been decreased.
- B** The activation energy has been increased.
- C** There are fewer successful collisions per unit time.
- D** There are more successful collisions per unit time.

- 5 An autocatalytic reaction is a reaction in which one of the products catalyses the reaction.

Which curve was obtained if the rate of reaction was plotted against time for an autocatalytic reaction?

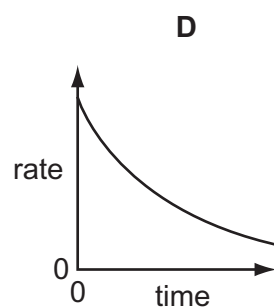
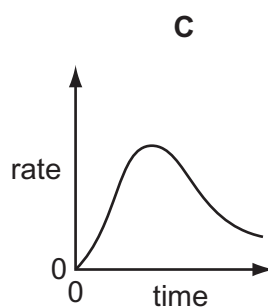
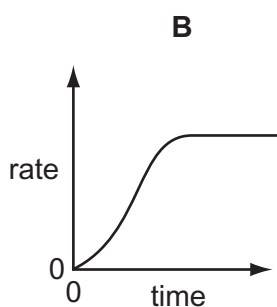
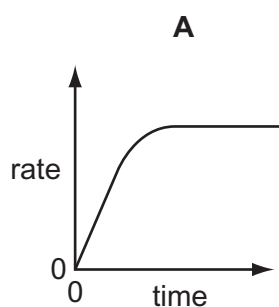


Question 7

9701_w13_qp_12 • Q5

- 5 An autocatalytic reaction is a reaction in which one of the products catalyses the reaction.

Which curve was obtained if the rate of reaction was plotted against time for an autocatalytic reaction?



- 2 Ammonia is manufactured by the Haber Process, in an exothermic reaction.

Assuming that the amount of catalyst remains constant, which change will **not** bring about an increase in the rate of the forward reaction?

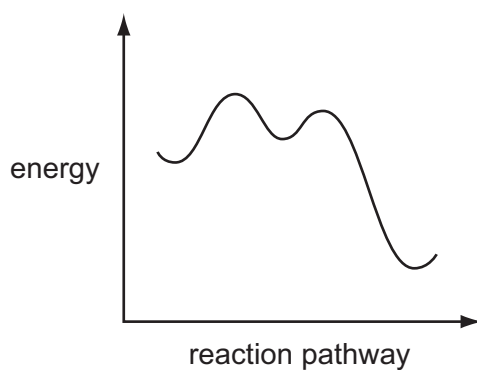
- A decreasing the size of the catalyst pieces
- B increasing the pressure
- C increasing the temperature
- D removing the ammonia as it is formed

- 1 Ammonia is manufactured by the Haber Process, in an exothermic reaction.

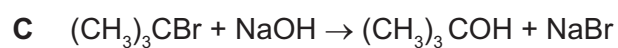
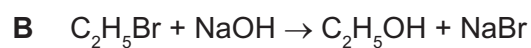
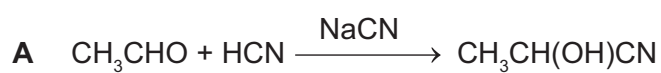
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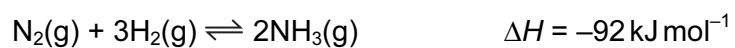
29 A reaction pathway diagram is shown.



Which reaction does **not** have such a profile?



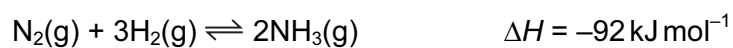
- 7 The Haber process for the manufacture of ammonia is represented by the following equation.



Which statement is correct about this reaction when the temperature is increased?

- A Both forward and backward rates increase.
- B The backward rate only increases.
- C The forward rate only increases.
- D There is no effect on the backward or forward rate.

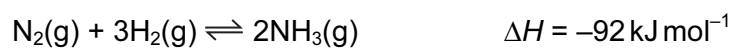
- 10 The Haber process for the manufacture of ammonia is represented by the following equation.



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- 8 The Haber process for the manufacture of ammonia is represented by the following equation.



Which statement is correct about this reaction when the temperature is increased?

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Answer Key

No.	Paper	Topic	Answer
1	9701_s23_qp_12	8. Reaction kinetics	C
2	9701_s22_qp_13	8. Reaction kinetics	D
3	9701_s21_qp_12	8. Reaction kinetics	D
4	9701_s19_qp_13	8. Reaction kinetics	D
5	9701_s18_qp_12	8. Reaction kinetics	B
6	9701_w13_qp_11	8. Reaction kinetics	C
7	9701_w13_qp_12	8. Reaction kinetics	C
8	9701_s12_qp_11	8. Reaction kinetics	D
9	9701_s12_qp_13	8. Reaction kinetics	D
10	9701_s12_qp_13	8. Reaction kinetics	B
11	9701_s10_qp_11	8. Reaction kinetics	A
12	9701_s10_qp_12	8. Reaction kinetics	A
13	9701_s10_qp_13	8. Reaction kinetics	A